

AWS EC2 – Amazon Linux Instance with Nginx using User Data (Detailed)

Task:

Launch an Amazon Linux EC2 instance and automatically install and configure Nginx using User Data script, then access the web page using public IP.

(a) Step 1: Login to AWS Console

1. Open browser and go to:
<https://aws.amazon.com/>
2. Click **Sign In**
3. Enter your AWS account credentials
4. Open **EC2 Dashboard** from Services menu

(b) Step 2: Launch EC2 Instance

1. Click **Launch Instance**
2. Enter Instance Name:
Example: NginxWebServer
3. Select Amazon Machine Image (AMI):
 - Choose **Amazon Linux 2 (64-bit)**
4. Select Instance Type:
 - Choose **t2.micro** (Free Tier eligible)
5. Create Key Pair:
 - Name: `mykey`
 - Type: **RSA**
 - Format: **.pem**
 - Download key and save it safely

(c) Step 3: Configure Network Settings

1. Allow Security Group rules:
 - SSH (Port 22) → for login
 - HTTP (Port 80) → for web access
2. Keep default VPC settings

(d) Step 4: Add User Data Script (IMPORTANT)

1. Scroll to **Advanced Details**
2. Find **User data** section
3. Paste the following script:

```
#!/bin/bash
```

```
# Update system packages  
yum update -y
```

```
# Install Nginx web server  
amazon-linux-extras install nginx1 -y
```

```
# Start Nginx service  
systemctl start nginx
```

```
# Enable Nginx on boot  
systemctl enable nginx
```

```
# Create custom web page  
echo "<h1>Welcome to EC2 Nginx Server (User Data Installed)</h1>" > /usr/share/nginx/html/index.
```